

Artificial Intelligence to Support the Training and Assessment of Professionals: A Systematic Literature Review

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Advances in Artificial Intelligence (AI) and sensors are significantly impacting multiple areas, including education and workplaces. Following the PRISMA methodology, this review explores the current status of using AI to support the training and assessment of professionals. We examined 83 research papers, analyzing (1) the targeted professionals, (2) the skills assessed, (3) the AI algorithms utilized, (4) the data and devices employed, (5) data fusion techniques utilized, (6) the architecture of the proposed platforms, (7) the management of ethics and privacy, and (8) validations of the proposals. The review highlights a trend in evaluating health-care professionals (especially surgeons) motivated by the critical role of hands-on training in these professions. Besides, the review reveals that data fusion techniques and certain technologies, like transfer learning and explainable AI, are not widely utilized despite their huge potential. Finally, the review underscores that most proposals remain within the research domain, lacking the integration and maturity needed for sustained use in real-world environments. Therefore, most of the proposals are not currently available to support the training of professionals. The insights of this review can guide researchers aiming to improve the training of professionals and, consequently, their education.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → **Artificial intelligence**; • **Human-centered computing** → **Human computer interaction (HCI)**; • **Social and professional topics**;

Additional Key Words and Phrases: Systematic review, artificial intelligence, data fusion, workforce training, professional competence

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1 Introduction

The advancement of Artificial Intelligence (AI) and sensors is transforming multiple domains, including education and workplaces [20, 86]. In education, AI is revolutionizing student assessments, offering more personalized and accurate evaluations [88]. AI can enrich educational experiences, for example, by providing adaptive learning platforms and real-time feedback [30]. On the other hand, integrating AI in work environments is changing work quality, increasing productivity, and optimizing resource management [86]. We can also combine AI, education, and work environments, employing AI to support the training and assessment of professionals [18, 99, 124]. Applying AI in the training and assessment of professionals is an opportunity to significantly elevate the workforce's skill set. This better education is especially relevant for workers whose lack of training might lead to fatal consequences [29, 70].

While AI can support theoretical and practical education, its most significant impact is observed in hands-on training. Many professionals suffer from a deficiency in hands-on training, often attributed to its prohibitive costs and scalability challenges [103, 111]. Furthermore, optimal training for certain professions, such as teaching, necessitates continuously repeating practical exercises [91]. The importance of addressing this gap is emphasized by the consequences that a lack of practical skills can produce, particularly in professions like first responders [7, 103, 111]. Besides technical skills specific to a particular profession, soft skills are also essential for the twenty-first-century workforce [75]. Soft skills are crucial to the performance and well-being of the workplace. Some soft skills are teamwork, communication, critical thinking, stress management, and emotional resilience.

Both practical and soft skills can be significantly enhanced through realistic exercises and simulations, which provide a safe and controlled environment for learners to practice and improve their skills [10, 51, 85]. These simulations are especially beneficial because they allow for increased frequency and adaptability of training sessions. Consequently, they enable learners to improve their skills through continuous practice. However, the effectiveness of these training exercises hinges on the quality of feedback provided to the learner. Without timely and accurate feedback, learners may repeat incorrect practices, developing bad habits [59, 109]. Continuous repetition of errors without appropriate feedback can be counterproductive, decreasing the student's learning. Using experts to provide immediate and comprehensive feedback can limit the flexibility of training sessions. Experts are not always available, and their time is a valuable resource that cannot be easily scaled. This handicap can significantly reduce the accessibility of the training, restricting its benefits [127].

To overcome the aforementioned problem, there is a growing interest in integrating advanced technologies and AI into training to complement or replace expert evaluation [34, 85]. AI can provide quick feedback to the learner, avoiding repeating bad habits. In addition, it solves the scalability problem and reduces the cost of the training [8, 100]. Besides, the AI assessment and feedback can be highly accurate, among other reasons, the high evolution of the sensor technologies in the last years. Sensors such as cameras, microphones, or accelerometers have decreased their price and increased their precision, making possible also the collection of data from multiple sources and processing of all of them combined through data fusion techniques and AI [3, 101].

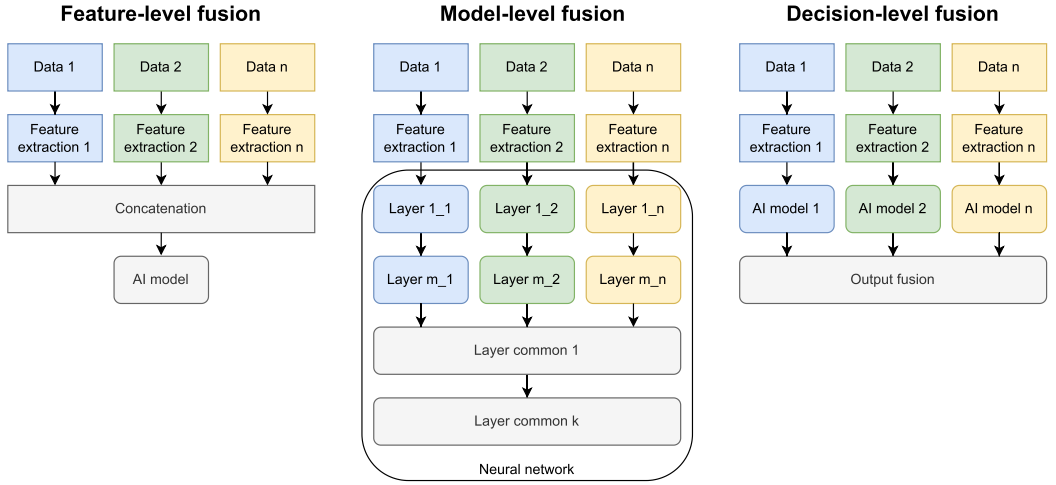


Fig. 1. Overview of the data fusion techniques.

Data fusion techniques integrate data from different sources and modalities to develop a more comprehensive analysis [119]. These techniques enhance the information provided to AI models, improving their accuracy and robustness [57]. The three primary data fusion techniques are explained below and depicted in Figure 1.

- **Feature-level or early fusion.** This method combines data by merging features at the initial processing stages. The data are concatenated before being introduced into the AI model, allowing the model to learn from the combined dataset.
- **Model-level or intermediate fusion.** This technique merges different datasets within the AI model itself. To implement, this technique often utilizes multi-head neural networks.
- **Decision-level or late fusion.** This approach processes each data source with its dedicated AI model. After making individual predictions, a final model or algorithm combines the predictions into a unified decision.

Despite the relevance of an updated analysis of AI for the training and assessment of professionals, to the best of our knowledge, no previous literature review has assessed its current status. Literature reviews on specific AI algorithms rarely include articles from both the educational and workplace fields. Besides, the research questions of these works are not focused on guiding novel researchers about the use of AI to support the training and assessment of professionals. Some examples are [44, 58]. In the educational context, some authors have previously analyzed the state-of-the-art of AI and education. However, these literature reviews mainly focus on analyzing tools for schools and not for the training of existing or future professionals [22, 24, 26]. From the educational context, the literature review that might intersect most with ours is [20]. Chen et al. [20] analyzed the application of AI on education in general, including some proposals that could be applied to learners of some professions. Other differences with our literature review are the lack of analysis of data fusion techniques and the adoption of the proposals. On the other hand, the literature reviews on AI and work environments primarily concentrate on how AI can enhance productivity and replace or alter certain professions [68, 86]. Combining professionals and education, there are limited examples. These few examples are tailored to particular professions and also neglect the review of the data fusion techniques. Yanik et al. [127] analyzed the use of deep learning to assess surgical skills, and Zhang et al. [131] reviewed the application of AI in medical

Table 1. Comparison of the Related Surveys to the Systematic Review Introduced in This Paper

Review	Educational objective	Application to professionals	Technologies evaluated	Data fusion analysis	Adoption analysis	Years analyzed
[58]	✗	✗	Auto machine learning	✗	✗	No limit-2022
[44]	✗	✗	Federated learning	✗	✗	2019–2023
[19]	✗	✗	Large language models	✗	✗	2020–2023
[24]	✓	✗	AI	✗	✗	2016–2022
[20]	✓	✗	AI	✗	✓	2010–2019
[22]	✓	✗	AI	✗	✓	2012–2021
[26]	✓	✗	AI	✗	✓	2016–2022
[79]	≈	≈ Surgeons	AI	✗	✓	2016–2020
[127]	✓	≈ Surgeons	Deep neural networks	✗	✗	No limit–2020
[131]	✓	≈ Healthcare	AI	✗	✓	1990–2022
[117]	✓	≈ Teachers	Virtual reality	✗	✓	No limit–2022
[95]	✓	≈ Teachers	AI and learning analytics	✗	✓	2017–2021
[68]	✗	✓	AI	✗	✓	No limit–2021
[86]	✗	✓	AI	✗	✓	1995–2020
Our review	✓	✓	AI	✓	✓	2018–2024

Legend: ✓ Yes – ✗ No – ≈ Partially.

This table highlights that no author has previously reviewed the use of AI to support the training of professionals and the importance of data fusion techniques has been neglected.

education. Similarly, Moglia et al. [79] explored the application of AI in robot-assisted surgery, identifying a significant number of papers focused on evaluating surgery skills. Wang and Li [117] examined the use of virtual reality in teacher education, but they neglected the application of AI. Salas-Pilco et al. [95] explored the use of AI and learning analytics in teacher education until 2021. Table 1 shows that no author has previously reviewed the utilization of AI to support the training of professionals, and the analysis of data fusion techniques has been neglected.

AI has a huge potential to improve the training and assessment of professionals, especially the need for practical training and soft skills. The high growth of AI and data fusion techniques makes necessary analysis of this field, identifying common approaches, gaps, and novelty ideas, serving as a guideline for novice researchers. However, the literature reviews on AI in education or AI in workplaces do not cover all the proposals to train and evaluate professionals and the reviews of specific professionals missed elements that could be reused for other professionals.

The rest of the paper is structured as follows. Section 2 describes the methodology followed in this systematic review, including the Research Questions (RQs), databases explored, search terms, inclusion and exclusion criteria, and review methods. Sections 3–5 present the analysis of the results and findings for the RQs. Then, the paper discusses the trends identified, the open challenges, and the main limitations in Section 6. Finally, Section 7 presents the main conclusions of the literature review.

2 Methodology

We followed a standard systematic literature review methodology, using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). We selected PRISMA to apply a validated methodology in the state of the art and allow other authors to replicate our results. We can see the flow diagram representing the stages of our systematic review in Figure 2. Our first step was to select meaningful RQs that needed answers in the state of the art. Then, we developed a set of queries to extract papers related to the application of AI for the training or assessment of professionals. However, some of the papers obtained from the queries were not related to the

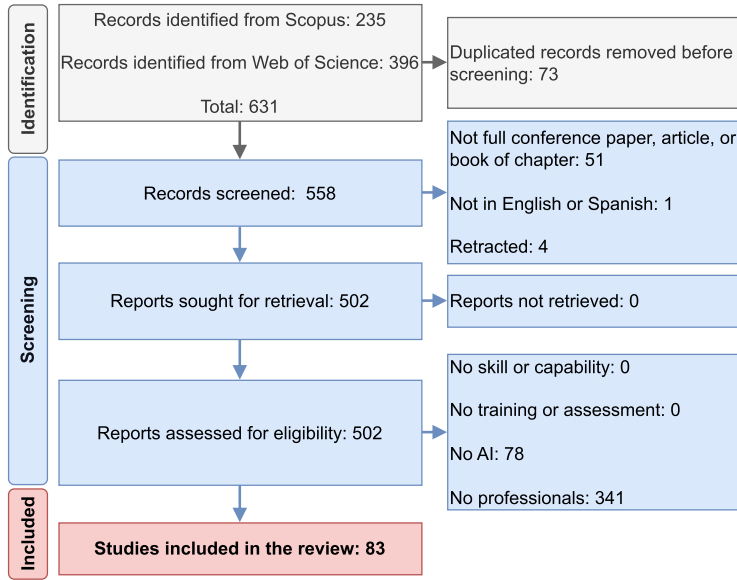


Fig. 2. Flow diagram of the PRISMA methodology followed to conduct the literature review.

objectives of our literature review; therefore, we included a screening phase to filter these papers. The inclusion criteria required that the papers had to train or evaluate skills or capabilities, include AI, and apply to professionals. Finally, we analyzed the papers, and we obtained the answers to our RQs. In this analysis, we followed an inductive coding scheme (also called open coding).

2.1 Research Questions

The objective of this literature review is to examine research papers related to the application of AI for the training and evaluation of professionals. How we addressed this objective through the defined RQ is depicted in Figure 3. In this topic, we deemed it relevant to gather information on the professionals who are the focus of current state-of-the-art tools and the skills measured. These two contextual questions (RQ1 and RQ2) aimed to identify neglected professionals and skills. Additionally, our analysis extended to the technological approaches, including the data employed, the technologies applied, and the data fusion techniques. These three technological questions (RQ3, RQ4, and RQ5) aimed to find innovative approaches and allow the readers to get inspiration to apply them to their research easily. Finally, we desired to analyze the final proposal of the authors, obtaining information about the tool's architecture, how the ethics and privacy were managed, and whether the authors validated the final systems with end-users. These last three questions on adoption and validation (RQ6, RQ7, and RQ8) were crucial for determining whether the tools were practically applied and tested or remained in the research field. These RQs are listed below:

- **Contextual aspects:**
 - RQ1. Which professionals are the proposals focus on?
 - RQ2. What skills/capabilities are trained or assessed?
- **Technological components:**
 - RQ3. Which AI technologies are applied?
 - RQ4. What types of data and devices are used?
 - RQ5. How much different data are collected, and what fusion techniques are applied?

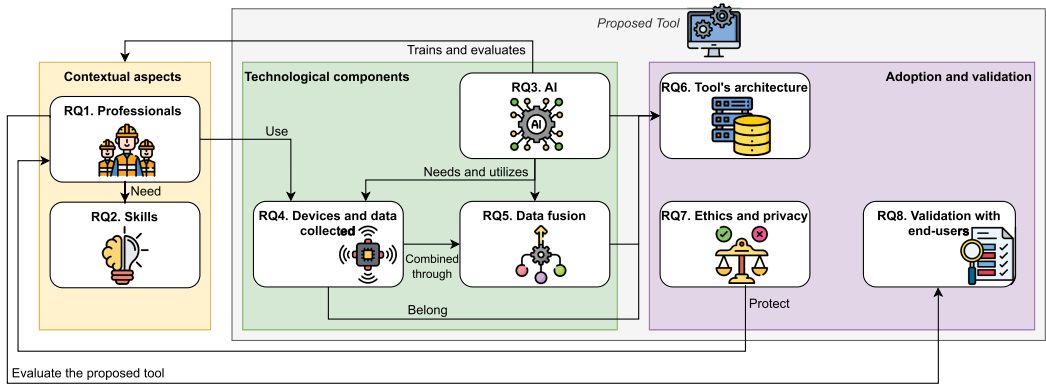


Fig. 3. Summary of the RQs addressed in this literature review and their relationships.

- **Adoption and validation:**

- **RQ6.** What is the architecture of the proposed tools?
- **RQ7.** How do the proposed tools manage ethics and privacy?
- **RQ8.** Is the proposed tool validated with end-users?

2.2 Identification

To answer the aforementioned RQs, we queried Scopus and Web of Science databases since they are the most widely used databases in scientific fields [17]. The objective of the review was to analyze publications that developed AI tools for the training or evaluation of professionals. Accordingly, we considered that the selected papers should satisfy the following conditions for the search in the databases:

- **Be focused on skills or capabilities.** This condition is fulfilled by adding this filter to the titles of the papers in the previous databases:
TITLE (skill OR competence OR capability OR capacity OR ability).
- **Train or evaluate the measured skills or capabilities.** To achieve this, we incorporated the following filter into the query:
TITLE (assessment OR training OR evaluation).
- **Include AI.** We excluded from this review all publications that did not involve the application or usage of AI. For this reason, we filtered the papers to ensure that at least one of these terms appeared in their title, abstract, or keywords: data fusion, multimodal, multi-modal, artificial intelligence, machine learning, or deep learning. We did not require that the title include all these terms because, in a preliminary analysis, we noticed that some authors did not incorporate them in the title or introduced only the name of the specific AI model they used. The filter to remove publications without AI was:
TITLE-ABS-KEY (data fusion OR multimodal OR multi-modal OR artificial intelligence OR machine learning OR deep learning).
- **Be applicable to professionals.** The tool should be suitable for professionals or workers. The problem with this condition is that some authors often ignore the term professionals and replace it for whom the tool is designed. In addition, this filter also eliminated generic proposals that could be applied to professionals. Instead, we included this criterion as part of the “inclusion/exclusion criteria” to be manually checked.

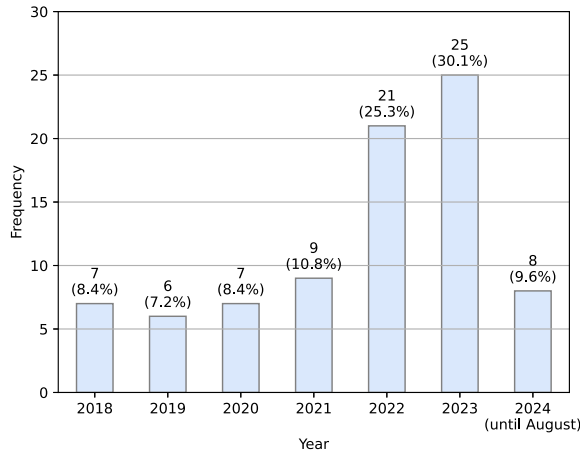


Fig. 4. Distribution of the studies analyzed per year of publication.

Considering the previous conditions, the final query employed in Scopus and Web of Science was:

TITLE ((skill OR competence OR capability OR capacity OR ability) AND (assessment OR training OR evaluation)) AND TITLE-ABS-KEY(data fusion OR multimodal OR multi-modal artificial intelligence OR machine learning OR deep learning).

We retrieved papers published only from 2018 to 2024. The query was performed on August 24th, 2024, and retrieved 235 papers from Scopus and 396 from Web of Science. After eliminating the duplicated papers, we had 558 to analyze.

2.3 Inclusion/Exclusion Criteria

In our first screening, we removed not full papers published in conference proceedings, journals, and book chapter papers to guarantee the quality of the publications analyzed. This quality screening eliminated a total of 51 papers. Then, we removed papers not written in English or Spanish (the languages in which the authors had high proficiency). This language filter only eliminated one paper in Chinese. We then eliminated four retracted articles. After this first screening, we had 502 articles, and we obtained the full papers of all of them. The second screening filtered papers not related to the objective of this literature review. Therefore, we eliminated 78 papers that did not include AI and 341 papers that did not apply to professionals. In this second screening, we did not find papers not related to measuring skills or capabilities for training or assessment. After this screening, all the papers applied AI to train or evaluate professionals.

2.4 Final Collection

The final collection analyzed contained 83 papers from 2018 to 2024. Figure 4 depicts the distribution of the analyzed papers based on their year of publication. In this figure, we appreciate an increase in the number of publications related to the topic reviewed from 2022 onwards. Due to the growing interest in this topic, this review has acquired higher relevance, especially to allow future researchers to understand it and find approaches they could reuse. On the other hand, Figure 5 shows that the 80.72% authors of this topic published their experiments through journal articles. The International Journal of Computer Assisted Radiology and Surgery was the journal most employed, presenting proposals to train surgeons.

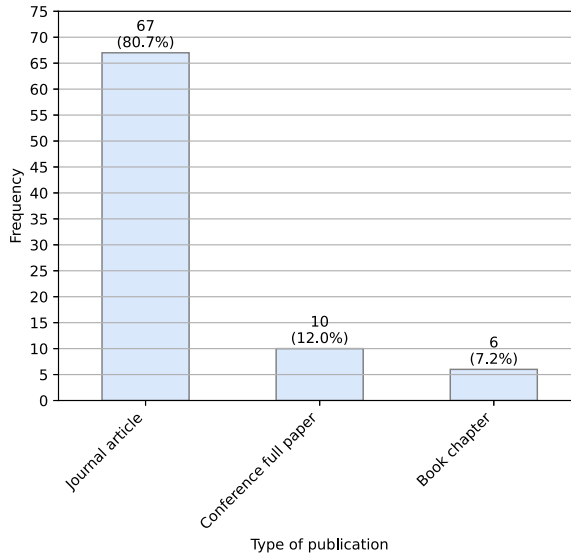


Fig. 5. Distribution of the studies based on their type of publication.

2.5 Review and Coding Process

In the coding phase of our review, we systematically gathered all the essential information required to address the aforementioned RQs. For this purpose, we adopted an inductive coding approach [1], directly deriving the codes from the qualitative data we were analyzing rather than predetermining them before the analysis. This coding process is inherently iterative and dynamic and has the flexibility to modify the coding structure to fit the emerging patterns and insights better. In our literature review, the initial analysis step was a preliminary examination of each paper. This step was crucial as it centered on meticulously extracting all essential information for coding each RQ. After this initial review, we developed a systematic and iterative process to code the information pertinent to our RQs effectively. This process was a recurring cycle of analysis, reflection, and refinement to ensure the maximum accuracy and relevance of our coding.

3 Professionals and Skills (RQ1 and RQ2)

We first identified the professionals targeted by the proposals (RQ1) and the competencies measured (RQ2). Table 2 summarizes the professionals and skills identified. This table shows that most of the proposals are focused on healthcare professionals (68.67%), mainly for surgeons (51.81%). The main reason is the fatal consequences of insufficient training of surgeons [45, 47, 61, 69, 105]. Traditional methods of practical training of surgeons are based on cadaver practice, which is expensive, time-consuming, and limits the frequent realization of this practical training. We also identified one evaluation of the surgery skills of veterinarians [66]. Furthermore, we found proposals to measure other healthcare skills other than surgery: sonography (2.41%, [64, 120]), psychotherapy (2.41%, [37, 110]), patient-physician communication (2.41%, [46, 63]), cannulation (1.20%, [73]), anesthetizing (1.20%, [97]), theoretical knowledge (1.20%, [78]), colonoscopy skill (1.20%, [122]), counseling (1.20%, [99]), chorionic villus sampling (1.20%, [56]), and hand-scaling (1.20%, [129]).

Educators have also received some attention in the gathered publications, with 7.23% of the publications. Of these publications, 6.02% focus on **English as a Second Language (ESL)** teachers, and only 1.20% (one paper) can be applied to any teacher. The papers focused on teaching ESL

Table 2. Papers and Rates Categorized According to the Professionals (RQ1) and Skills (RQ2) Targeted

Professionals	Skills	Frequency	Works
Healthcare professionals	Surgery	43 (51.81%)	[8, 11, 14, 15, 25, 27–29, 32, 33] [34–36, 38, 45, 47–51] [53–55, 60, 61, 65, 69, 70, 76, 84] [94, 96, 100, 102–105, 108, 112, 113] [121, 130, 134]
	Sonography	2 (2.41%)	[64, 120]
	Psychotherapy	2 (2.41%)	[37, 110]
	Patient-physician communication	2 (2.41%)	[46, 63]
	Cannulation	1 (1.20%)	[73]
	Anesthetizing	1 (1.20%)	[97]
	Theoretical knowledge	1 (1.20%)	[78]
	Colonoscopy skill	1 (1.20%)	[122]
	Counseling	1 (1.20%)	[99]
	Veterinary surgery	1 (1.20%)	[66]
	Chorionic villus sampling	1 (1.20%)	[56]
	Hand-scaling	1 (1.20%)	[129]
Any professional	Communication	5 (6.02%)	[71, 85, 90, 109, 123]
	Collaborative problem solving	2 (2.41%)	[10, 23]
	Leadership	1 (1.20%)	[42]
	Oral presentation	1 (1.20%)	[82]
	English expression	1 (1.20%)	[118]
	English translation	1 (1.20%)	[133]
Athletes	General sports movements	2 (2.41%)	[21, 128]
	Balance control	2 (2.41%)	[124, 125]
	Basketball	1 (1.20%)	[80]
	Table tennis	1 (1.20%)	[98]
	Ski	1 (1.20%)	[83]
Teachers	English teaching	5 (6.02%)	[72, 77, 89, 106, 126]
	General teaching	1 (1.20%)	[111]
Militaries	Firing	1 (1.20%)	[18]
Musician	Piano	1 (1.20%)	[132]

include in their platform different constraints that do not allow the application to other teachers [72, 77, 89, 106, 126]. In contrast, Tang et al. [111] focus on microteaching skills applicable to any teacher. Similarly, 8.43% of the publications focus on skills related to sports, 4.82% can be applied to any athlete [21, 124, 125, 128], and 2.40% are focused on specific sports (basketball [80] and skiing [83]). To develop platforms to support the training of any athlete, there are two approaches in the state of art: measuring skills useful for any athlete, such as balance control [124, 125] or measuring the accuracy of any movement and posture [21, 128].

We also identified that 13.25% of the papers measured generic skills useful for any professional. These skills are communication [71, 85, 90, 109, 123], collaborative problem-solving [10, 23], leadership [42], oral presentation [82], general English expression [118], and translation [133]. One of the benefits of these studies is their broad applicability, offering insights that can be utilized across most professional fields. However, in contrast to these broadly applicable studies, two publications were destined for too specific professionals: one examined skills specific to piano musicians [132], and the other assessed the firing abilities of military personnel [18].

4 Technological Components (RQ3, RQ4, and RQ5)

This section analyzes the AI algorithms, the processed data, and the data fusion techniques employed in the reviewed papers. This section presents what AI models and how they have been used to facilitate their application for other researchers.

4.1 AI Technologies Employed (RQ3)

In our literature review, we meticulously compiled the various AI technologies utilized in each reviewed paper, and we summarize the results in Figure 6. It is noteworthy that machine learning techniques predominate, being employed in the majority of the papers. Among the machine learning techniques, the authors of the analyzed papers employed supervised, self-supervised, unsupervised models, and transfer learning. Supervised learning is the most dominant machine learning approach, employing ensemble models (13.25%, [65, 70, 102, 109]), supervised deep learning models (50.60%, [21, 66, 98, 124]), and traditional supervised models (39.76%, [100, 113, 129, 134]). This predominance is primarily attributed to the common practice in the papers analyzed of acquiring a labeled dataset pertinent to the target task and training a supervised model on this dataset. The review also recognized two tendencies in the unsupervised models focused on clustering tasks. The first trend is hard clustering, where each element is exclusively assigned to a single cluster [10, 23, 105]. The second trend is fuzzy clustering, where elements possess a degree of membership to each cluster [72, 77, 89, 126]. Transfer learning, though less common, emerges in two forms: basic transfer learning [130] and auxiliary supervision [50]. In auxiliary supervision, the model is pre-trained in one task related to the target task, and the pre-trained model is then trained in the final task. This technique allows the authors to reduce the data required to train the AI model. In the review context, this approach is highly beneficial, because the authors could pre-train the model with public datasets, and then they only need to collect a small dataset of the targeted task. Lastly, the review detected a paper where the authors employed self-supervised learning. Wang et al. [121] labeled data from common surgical training tasks to jointly train a model with both labeled and unlabeled data. This is another interesting approach to reducing the labeled data needed to train a machine learning model. Misra and Suresh [78] utilized ChatGPT, a large language model, to provide automatic assessments in medical education. Among other techniques, ChatGPT was developed using self-supervised learning.

To improve the explainability of the machine learning models, two papers explored post-hoc eXplainable Artificial Intelligence (XAI) [36, 123]. Post-hoc XAI consists of adding to a machine learning model another algorithm to understand the reasoning followed by the model. These two articles include interesting proposals, as it would be very valuable to add to the skills assessment of why a value is predicted, allowing validation by experts and generating feedback with recommendations. Computer vision is present in 51.81% of the papers, mainly focused on measuring the subject's movements non-invasively through cameras. Different Natural Language Processing (NLP) techniques were also employed in 14.46% of the papers to analyze audio transcription or the subject's responses. Besides, it is worth noting that 14.46% employed deep learning models to apply feature extraction from images or videos. For example, to obtain the position of the surgeon's instruments. Furthermore, 6.02% of the papers employed binary rules [10, 35, 42, 46, 82] and another 4.82% fuzzy rule-based systems [18, 33, 34, 49]. Rule-based systems are an expensible alternative to machine learning. Finally, Iranfar et al. [54] stand out for developing an expert system to evaluate surgical skills based on Dempster-Shafer Theory. Dempster-Shafer Theory offers a mathematical framework to represent and reason with uncertainty, allowing the integration of imprecise and inconsistent data.

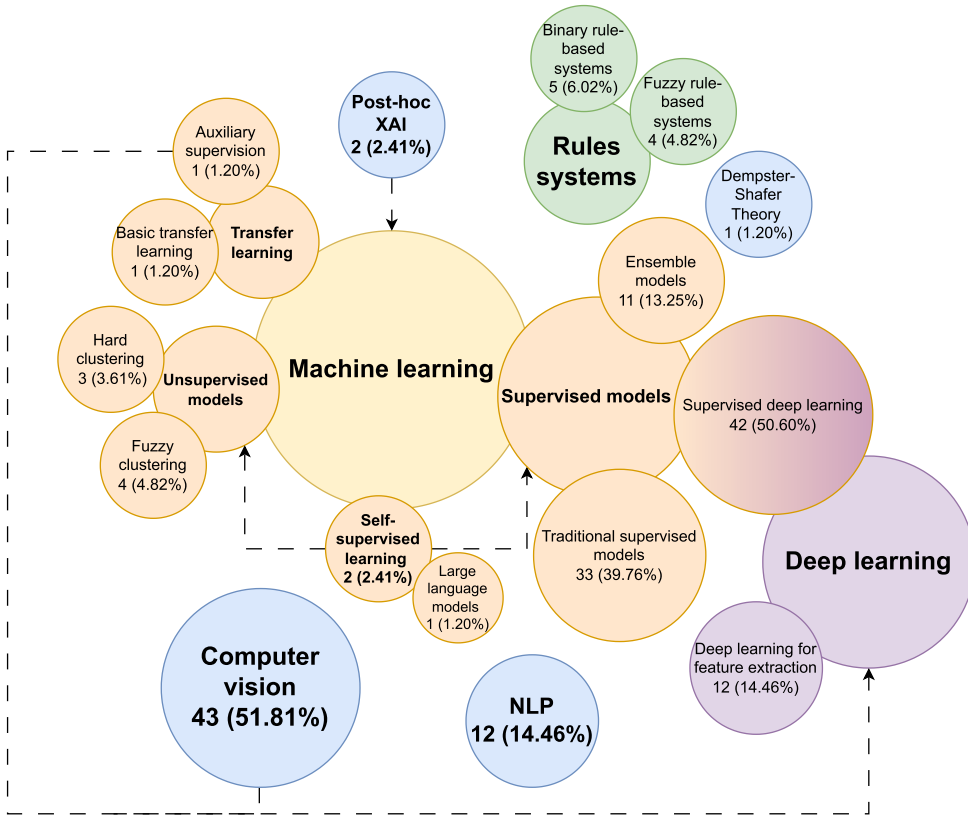


Fig. 6. Utilization frequency and rates of the AI techniques across the analyzed studies (RQ3). Lines represent relations/similarities between two techniques, and the size of the bubbles illustrates the frequencies.

4.2 Data and Devices Analyzed (RQ4)

The analysis of data and the devices employed to collect them are summarized in Figure 7. In this figure, it is important to clarify that one manuscript could use multiple data and devices. The figure shows that audio-visual data are the most common, especially video recordings obtained by cameras that are employed in 50.60% of the papers. The main advantage is that cameras are non-invasive devices and low-cost sensors. However, to obtain a higher precision of the position and movement of objects, Ebina et al. [28, 29], Yao and Li [128] employed infrared cameras. Audio is another non-invasive and low-cost data source that allows the processing of acoustic and linguistic features through audio transcription. 13.25% of the analyzed papers employed microphones to collect audio (for example, [37, 46, 82, 85]). On the other hand, biometric data are only employed in five papers [27, 66, 100, 102, 108]; some reasons could be the high price, and the invasiveness, thus, uncomfortable for sustained use. The biometric signals observed are brain signals, electromyography (EMG), and force myography (FMG). Brain signals are collected through Brain-Computer Interfaces (BCI) and contain the brain's electrical activity [27, 108]. EMG is a biometric signal that measures the electrical currents generated in muscles during their contraction [66, 102]. Finally, FMG is an alternative to EMG that measures the muscle force during their contraction [100].

The movement of users is also measured through biometric movement (6.02%), eye tracking (2.41%), data from kinematic instruments (22.89%), pressure (9.64%), sociometric (1.20%), and

virtual reality data (3.61%). The biometric movement contains the movement of one part of the body through an accelerometer placed in the measured part [85, 98, 102, 128, 134]. In contrast to video and infrared video recordings, the movement is measured directly by the devices without image preprocessing. Eye trackers are non-invasive devices that combine multiple cameras and infrared sensors to measure only the eye's movement and/or what is paying attention to [63, 65]. Cameras and the required preprocessing can also be used to collect eye tracking, but eye trackers reduce the preprocessing steps and usually improve the precision. The category of Kinematic instrument data generalizes the position and the movement of kinematic devices. In this literature review, many papers focused on measuring surgery using these devices, such as [14, 15, 51, 104, 121]. Similar to biometric movement and eye tracking, the movement of the instruments can be measured through cameras that are less expensive devices but require a bigger preprocessing [35, 38, 60, 70, 84, 113]. In the articles reviewed, pressure sensors were used to collect information about the force applied by a human to an object or a surface. For example, these sensors were utilized to quantify the force exerted through legs to measure balance control skills [124]. Additionally, sociometric data represent a category of data that is relevant for analyzing communication tasks. This category comprises various features, such as the total duration of speech, vocal volume, pitch, and postural mirroring in interactions with others. While cameras and microphones can capture some of these features, sociometric devices capture and process the required data to provide these features directly [85].

Virtual reality is another approach to train and assess professionals through simulations. Virtual reality sensors gather user interaction information (3.61%), commonly through remote controllers [50, 94, 121]. Besides the sensors, virtual reality employs a display (6.02%) to make the simulations more realistic [25, 50, 83, 94, 121]. The display does not require virtual reality sensors, and the authors can incorporate additional sensors [25, 83].

Text is also a data source that comprises highly relevant linguistic features. In the analyzed papers, text is collected from platforms that gather the user's text answers (6.02%) [10, 42, 78, 110, 133], obtained through audio transcriptions [37, 46, 90, 109, 111, 123] (7.23%), and presentation slides [82] (1.20%). Notably, 11 papers collected audio (13.25%), but only the six aforementioned manuscripts extracted linguistic features through audio transcriptions. Ochoa and Dominguez [82] excel because they analyzed slide quality, considering the text size, length, and color contrast to improve presentation skills.

Log data is a miscellaneous category that comprises information from the user's interaction with software platforms. Logs are collected in 12.04% of the studies [10, 23, 42, 110]. The last type of data identified is experience data, Chandan et al. [18] collected information about the subjects' experience, such as service length and number of firing practices.

4.3 Data Fusion Analysis (RQ5)

The number of data employed in the reviewed papers is summarized in Figure 8. We identified the data analyzed simultaneously and how many of these data are from different sources. In this RQ, we have distinguished between data stream and data source. A data stream refers to data gathered from a data source. From one data source, we could obtain multiple data streams. For example, Subedi et al. [108] collected six brain signals (data streams), but they were all from the same source. The results of RQ5 show that 74.70% of the papers employed only one data stream, and 79.52% utilized data from a single source. Only 15.66% and 6.02% of the papers analyzed two and three data streams, respectively. The papers that studied the higher number of data streams are [100, 108], which employed six and eight biometric signals, but all of them were from the same source. Pereira et al. [85] are the authors who analyzed data from more sources, collecting audio, video, sociometric data, and movement collected by an accelerometer. We have also analyzed the change in the number of data streams and sources by the publication year. We applied the Shapiro-Wilk

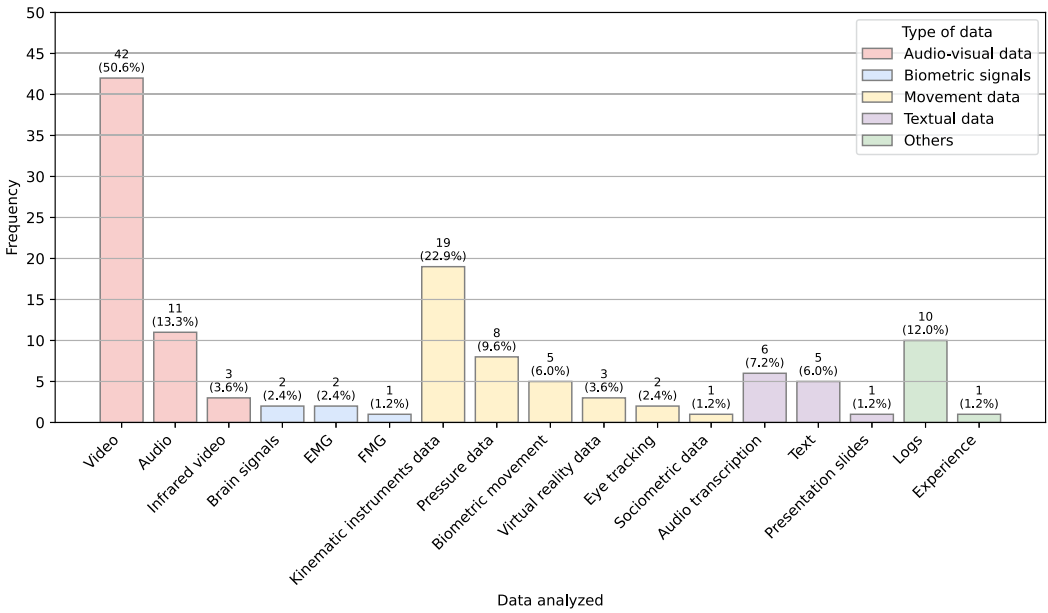


Fig. 7. Frequency of the different data types utilized in the papers examined (RQ4).

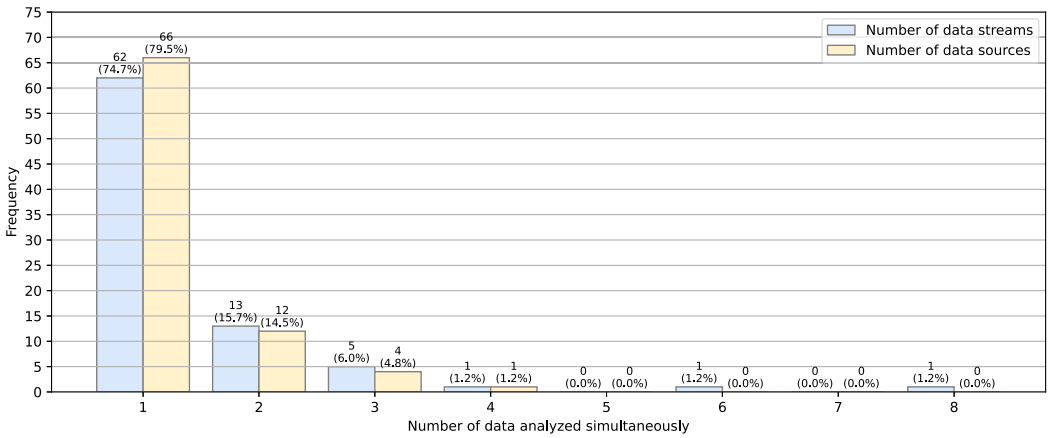


Fig. 8. Distribution of the papers based on the number of data analyzed (RQ5).

statistical test, and in both cases, we could not assume that the data follows a normal distribution. Consequently, to identify statistical differences, we employed the Kruskal–Wallis statistical test. However, both tests did not reveal any significant differences. The P-values obtained are 0.51 for analyzing the data sources and 0.83 for data streams by the year of publication. Consequently, the number of data sources and streams has not significantly changed over the years.

We complemented the previous analysis by exploring the data fusion techniques utilized, summarized in Figure 9. The most common approach identified involves using a single data stream without data fusion. This approach is employed in 68.67% of the papers, which is the expected conclusion considering the few data streams analyzed by the authors (Figure 8). Another approach includes analyses that involve two or more independent data streams without fusion.

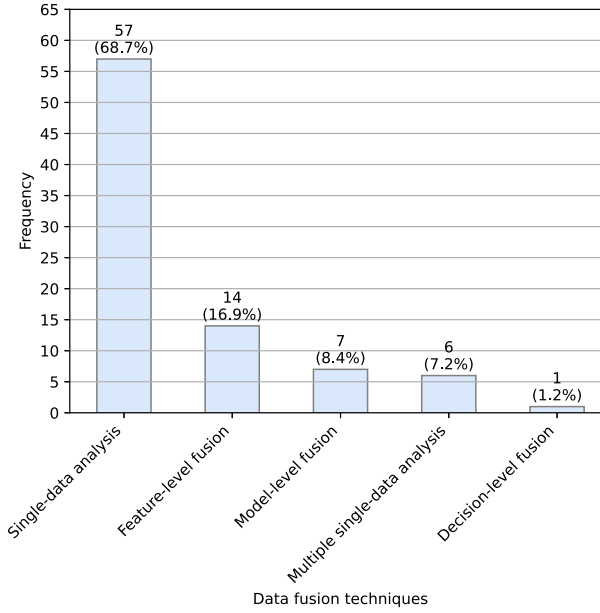


Fig. 9. Frequency of each data fusion technique in the analyzed papers (RQ5).

This technique was utilized in 7.23% of the papers [37, 42, 46, 63, 82, 111]. Finally, data fusion techniques are quite uncommon. Feature-level fusion was applied in 16.87% of the papers, and model-level fusion in 8.43%. However, only Rasipuram and Jayagopi [90] employed decision-level fusion. These authors are also the only ones who tested different data fusion techniques, comparing the results of feature-level and decision-level fusion.

5 Tool's Architecture and Testing (RQ6, RQ7, and RQ8)

In Table 3, we summarize the type of architecture of the presented tools. The lower level of maturity is the lack of architecture, which does not include any prototype for applying the author's proposals. 30.12% of the papers do not have an architecture for the proposals. Furthermore, these works did not collect their own datasets or include any prototypes. This is common in the papers focused on measuring surgical skills, especially using the dataset JHU-ISI Gesture and Skill Assessment Working Set (JIGSAWS) [11, 14, 32, 38, 55, 103]. 51.81% of the papers presented an architecture for individual assessments encompassing where the components were not fully automatized and incorporated into one platform. This category is common in papers where the authors developed their own case study and AI models but did not develop a fully integrated platform combining the data collection, processing, and access to the results. For example, Guo et al. [45] conducted a case study to collect the necessary data for training and assessing supervised AI models to evaluate surgical skills. Although the authors developed the case study and AI models, they did not integrate them into a final platform. In contrast, 3.61% of the authors developed final applications in individual computers [8, 83, 94]. Similarly, 14.46% of the analyzed papers developed a final application running on a server and typically accessed through a web browser. We have also analyzed the change in architectures by the publication year through the Chi-square statistical test. The P-value was 0.07, indicating no significant change in the architectures over the years.

The previous analysis shows a lack of maturity in the proposals, mainly only focused on research purposes. This finding is also supported by the ethics and privacy analyses developed to

Table 3. Papers and Rates of the Architectures used in the Proposed Tools (RQ6)

Architecture	Frequency	Works
Architecture for individual assessments	43 (51.81%)	[10, 21, 25, 27–29, 33–35, 45] [47–51, 60, 61, 65, 66, 69] [70, 73, 76, 80, 85, 90, 97, 98, 100, 102] [105, 108, 111–113, 118, 120, 121, 124, 125] [129, 130, 134]
No architecture	25 (30.12%)	[11, 14, 15, 32, 36, 38, 53–56] [63, 64, 71, 72, 77, 78, 84, 89, 96, 103] [104, 106, 123, 126, 132]
Central server	12 (14.46%)	[18, 23, 37, 42, 46, 82, 99, 109, 110, 122, 128, 133]
Individual application in a computer	3 (3.61%)	[8, 83, 94]

Table 4. Papers and Rates of the Evaluations of the Proposals Presented in the Examined Publications (RQ8)

System validation	Frequency	Works
No validation	68 (81.93%)	[10, 11, 14, 15, 18, 21, 23, 25, 27, 28] [29, 32–36, 38, 45, 47, 48] [49–51, 53–56, 60, 61, 64] [65, 66, 69–73, 76–78] [80, 84, 85, 89, 94, 96–98, 102, 103] [104–106, 108, 112, 113, 118, 120, 121, 123] [124–126, 128–130, 132, 134]
System validation with end-users	12 (14.46%)	[37, 42, 46, 63, 82, 83, 90, 99, 109, 111, 122, 133]
Data collection feedback	2 (2.41%)	[8, 100]
System validation with non-end-users	1 (1.20%)	[110]

answer RQ7. These analyses revealed a critical problem: none of the papers reviewed had analyzed ethical and privacy management in their proposed platforms. One of the main causes is that most proposals are far from being applied in real-world environments.

The validation of the proposed platforms is summarized in Table 4. 81.93% of the proposals were not tested. Besides, two papers only collected some feedback about the case study developed to gather the data employed by the authors [8, 100], but they did not test the final proposal. Tanana et al. [110] tested their proposal but did not involve the final professionals for whom the platform was destined. Finally, 12 papers included an extra case study to assess the final training platforms presented using the end professionals. These results are also supported by Figure 10. This figure shows that most papers only developed one case study to collect data to develop their AI models, and 13 papers analyzed the final platform.

6 Discussion

In this section, we discuss the main findings, open challenges, and limitations of the review. Figure 11 summarizes these principal findings.

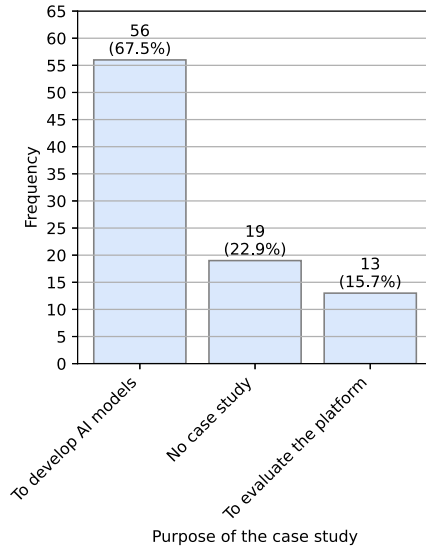


Fig. 10. Objectives of the case studies developed in the papers and their frequency (RQ8).

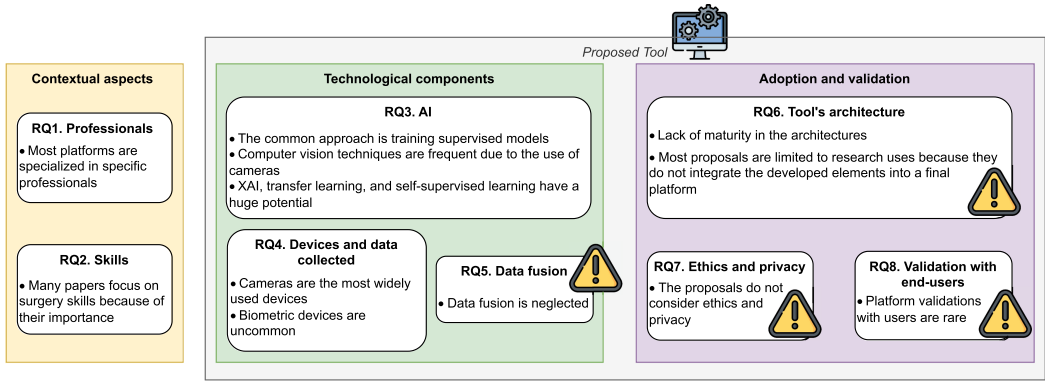


Fig. 11. Summary of the main findings identified in each RQ.

6.1 Current Trends

First of all, we analyzed the professionals targeted by the proposed platforms and the skills measured to answer RQ1 and RQ2. These analyses revealed that certain studies focused on particular skills and professional groups. These groups include healthcare professionals [73], educators [106], athletes [80], military personnel [18], and musicians [132]. It was noted that some proposals were highly specialized, limiting their applicability to other professionals within the same field; for example, the tools designed to train only ESL teachers [89, 106] or basketball players [80]. Within these proposals for training specific professionals, there are a large number of studies on training healthcare professionals [37, 97]. Two main reasons are that the lack of training of these professionals can have major consequences and that they are difficult to train practically. The most common example is the proposals designed for surgeons, with 51.81% of the works intended for them [84, 103]. Other proposals can be applied to professionals in the same field, for example, the tools that evaluated general sports movements [124, 125]. Finally, a few papers (only 13.25%)

focused on generic skills useful for any professionals, such as communication [110], collaborative problem-solving [85], leadership [42], oral presentation [82], and English expression [118].

The analysis of the AI technologies for RQ3 identified that machine learning, especially supervised machine learning, is the typical approach [37, 65, 106] to evaluate skills. Authors commonly collected their own dataset or employed a public dataset to train and test their supervised models. In contrast, rule-based systems and unsupervised models were rarely employed [10, 18, 23, 33] even though they do not require a large number of practitioners. It is also worth noting the lack of proposals that considered transfer learning [130], auxiliary supervision [50], and self-supervised learning [78, 121], which decrease the amount of data necessary to train the models. Reducing the data needed to develop the AI models is an important factor because getting practitioners as volunteers and experts to label the dataset is challenging [127]. This finding is also more relevant because supervised deep learning requires high amounts of data for training, and it is employed in 50.60% of the papers. Post-hoc XAI, despite its significant value in this context, is also neglected, with only two studies employing it [36, 123]. Thanks to the post-hoc XAI, users would be able to understand why the model considers high and low performance, and also the tool would be able to generate feedback and recommendations to improve their skills [40]. In addition, only Misra and Suresh [78] used generative AI, despite its high performance in NLP tasks and explainability [93]. The models developed by the authors could offer various performance metrics, which could be integrated into the generative model to provide comprehensive feedback with recommendations. For example, a platform integrating these models could evaluate the performance of a surgeon as poor due to imprecise instrument movements and suggest changes in how the surgeon holds the instrument to improve the precision. This feedback can also support educators in verifying the accurate automatic evaluation.

The analysis of data and devices employed in the analyzed papers (RQ4) shows that cameras are highly used because they are low-cost and non-invasive devices that can evaluate numerous skills. Cameras are easy to integrate into any work environment and simulations without hindering the users' movement. However, the videos commonly required a feature extraction to gather meaningful features. The typical use of cameras is associated with the frequent application of computer vision techniques and deep learning models. Movement sensors allow us to skip this preprocessing; however, they are more invasive and, therefore, less adequate for sustained use. This issue is mitigated when assessing a particular skill requires the user to employ an object, like a surgical instrument, which may incorporate movement sensors [14, 36]. Microphones are the only non-invasive and low-cost sensors that are useful to measure speech-related skills, for example, teaching [111]. Nevertheless, some skills, such as surgery, cannot be measured through audio. We also detected that many authors overlooked extracting audio transcriptions to obtain linguistic features. Biometric sensors are only employed in five papers [27, 66, 100, 102, 108]; one possible reason could be that they are expensive and research-oriented devices, not suitable for continuous or long-term use [6]. It was surprising to find that no author has employed heart rate, since it can be measured through cheap smartwatches. Finally, collecting data on user interactions with platforms is also interesting, especially if these platforms offer simulations of situations in which the user would be involved during his working day [107]. These simulations could be enhanced by virtual reality.

For RQ5, we delve into the quantity of data analyzed simultaneously by the authors and the specific data fusion techniques employed in their research. The findings from both analyses pointed out that the researchers of the examined articles highly neglected data fusion. One reason could be that the datasets used typically do not include more than 20 users, it may not be common to use many different data as this requires collecting more subjects to train supervised AI models. This is especially noteworthy because the JIGSAWS database includes kinematic and video data, but some

authors only use one, for example Benmansour et al. [14], Jian et al. [55], Kitaguchi et al. [61], Soleymani et al. [103]. Furthermore, expanding data volume can lead to more complex AI models, which in turn need extended training periods [9]. As the amount of data analyzed grows, the time needed to deliver feedback also rises due to the extension of the processing required. This is particularly crucial for real-time feedback, which is especially relevant for identifying wrong behaviors, increasing motivation, and allowing training adaptation based on the feedback generated. Nevertheless, real-time feedback with multiple data sources demands more complex architectures for the final systems. Finally, the increase in sensors and computational resources affects the cost of the final proposal, which could lead to a reduction in the users that have access to the final system [119].

RQ6, RQ7, and RQ8 identified a worrying trend: a lack of maturity of the proposed tools. 30.12% of the proposals do not have any type of architecture, being only isolated experiments. Besides, another 51.81% of the proposals do not automate and combine all the modules in one final tool, making it unsuitable for real-world environments. This lack of maturity leads authors to not consider ethical and privacy issues, and only 12 studies validated the platform with end-users. These findings are also related to the usual approach followed by the majority of the authors developing AI models to measure skills, leaving the proposed tools in the field of research [14, 38, 55]. One reason for the lack of maturity is the need to implement software that integrates all the developed components. Designing and developing this software can be time-consuming, particularly integrating the recording devices into the final platform. However, software development is a well-established field where it has already been demonstrated that these and other major problems can be addressed [4]. It should also be noted that many of the databases used by the authors are very small, among other reasons, due to the difficulty of getting a large number of different users [7]. For example, the JIGSAWS database only contains data from eight different users [14]. However, although most of the proposals are limited to research applications, the architecture, privacy, and ethics must be considered even for research applications. If these aspects are not considered, the proposals might be far removed from reality and may not be viable for use. Furthermore, involving end-users in the design phases is essential to guarantee that the platforms suit these professionals [16].

In summary, this review identified significant interest in evaluating healthcare professionals, particularly surgeons, due to the critical importance of their training. It also revealed that most authors overlooked data fusion techniques and advanced AI technologies like transfer learning and explainable AI. Finally, this literature review highlights the lack of maturity of most proposals. The majority of the authors did not integrate the developed AI models into the final platforms. Consequently, these proposals remain unvalidated and are not available to support the training of professionals. Figure 11 summarizes the aforementioned main findings.

6.2 Open Challenges

We have identified a lack of proposals for many types of professionals because the majority were designed for healthcare professionals. Many authors focus on evaluating surgeons as their performance is critical to the survival of patients. Other factors to consider are how expensive or unsafe is the alternative training or evaluation, the scarcity of these professionals, and the time an expert needs to do manual evaluations. Furthermore, the prevalence of professionals in a field who are also engaged in research is influential; fields like medicine have a substantial number of researchers and receive considerable funding. However, this gap affects many professionals, and errors can lead to deadly consequences. For example, first responders make critical decisions in stressful situations. Therefore, the training of these professionals could be improved through practice exercises that simulate stressful conditions [7]. The development of more generic proposals is an open challenge in the field. For example, instead of evaluating ESL teachers, abstract to teachers in general. Other work to be addressed is to study the transference of tools for training one type

of professional to another and what techniques could be applied. The focus on extremely specific skills of certain professionals also produces a lack of proposals for evaluating soft skills useful for all professionals, such as teamwork, communication, critical thinking, or emotional resilience [82, 85, 118]. Therefore, future researchers may explore other professions, such as first responders. They can also assess the transfer of AI models between work environments or develop generic platforms that assess soft skills.

The analysis of the AI technologies employed in the reviewed papers showed promising alternatives to supervised AI models for assessing skills, such as rule-based systems [10, 34, 82]. These AI methods allow automation of the manual evaluation used by experts. However, transferring expert knowledge into a rule-based system is a significant challenge because it requires a deep understanding of the domain and the ability to translate this expertise into rules for the AI system. Furthermore, this technique also demands the validation of the rules from multiple experts to avoid bias caused by the perspectives of a limited number of experts. Despite these challenges, rule-based systems have a huge potential thanks to their explainability [43, 115]. Post-hoc XAI addresses the explainability of the evaluations provided by black-box models [36, 123]. Explainability is important in the reviewed context because it allows evaluators to know what causes the high or low performance of the professionals, increasing the trustworthiness of the evaluations and enabling informed interventions. Explainability is especially useful when these systems are embedded within courses with also formative content, allowing the instructor to adapt the formation based on the students' needs. The evaluated professionals also benefit from explainability because they know what to improve and train for. Besides, explainable models open the possibility of including personalized feedback and recommendations. The generation of feedback is also an open issue neglected by many authors. Most papers only provide an evaluation score but not compressive feedback, including the mistakes and suggestions for the professionals. Without these suggestions, the professionals' learning capacity through the proposed tools is highly reduced [12]. Another pending challenge is using generative AI to improve the feedback shown to professionals, providing personalized recommendations and supplementary materials [87]. We encourage future researchers to explore AI models other than supervised machine learning models. Especially interesting are solutions that allow to improve the explainability of the predictions.

During the investigation of the data collected and the devices utilized, cameras showed a high potential to train and assess professionals because they are non-invasive and low-cost. However, other devices could be explored, such as infrared cameras, to improve the accuracy of the evaluations. It is also worth noting that some skills, such as sports, require many movements. Therefore, other researchers could analyze how to improve the accuracy of the systems with additional cameras in different positions. Besides, photoplethysmogram sensors are available in low-cost wearables to measure heart rate, for example, for stress management evaluation [5]. Including position and accelerometer sensors in the instruments used by the professionals is also an alternative to cameras [38, 50, 55]. In addition to tracking the surgeon's instruments, they might be used to assess painters, drivers, and chemists, among others. Future researchers could analyze the ease of integrating these devices into real tools, how user-friendly they are, and compare the accuracy of the evaluations using different sensors. Furthermore, it is necessary to evaluate how well these sensors are adopted by the professionals they are intended for. Ensuring that these professionals feel comfortable with the devices is crucial to avoid causing stress and added complexity in the training [116].

Virtual reality has noteworthy potential to create immersive simulations that enhance the realism of hands-on training. By offering a safe and controlled environment, virtual reality enables professionals to practice complex procedures and develop decision-making skills without the threat of real-world consequences [62, 74]. Other authors have already demonstrated the advantages of

this technology in educational contexts [52, 114]. However, only five of the papers analyzed in this review employed it [25, 50, 83, 94, 121]. Therefore, integrating virtual reality in the training or assessment of professionals supported by AI is an area that requires attention from future researchers. Another necessary research is to compare the realism of these virtual environments for different groups of professionals. The usefulness and effectiveness of virtual reality might vary depending on the virtualized work environment. Additionally, it is necessary to assess whether there are differences in the evaluations and performance of the professionals in simulated versus real-world settings. Therefore, we suggest that future researchers explore new devices to improve the accuracy of the evaluations and the realism of simulations.

We have identified a significant gap in the study of the number of data streams analyzed simultaneously and the data fusion techniques employed. Despite the high potential of multimodal analysis and data fusion techniques, 74.70% of the papers used only one data stream. One reason might be that supervised models, the most common AI models applied by the authors, require larger datasets as input data and training time increase. Therefore, collecting larger datasets with more data sources and users to test different data fusion techniques is also an open research area that can be addressed in future research. We have also evaluated the amount of data used over the years, and no significant differences exist. Over time, data collection devices have become more affordable, and computing resources have increased and become cheaper [13, 39]. However, this improvement in available resources has not increased the amount of data collected and analyzed in the papers reviewed.

Therefore, the analysis of the technological questions (RQ3, RQ4, and RQ5) identified that advanced AI technologies are not widely employed despite their potential to enhance the accuracy and interpretability of AI models. This represents a missed opportunity to improve the proposed training and assessment platforms. Data fusion techniques enable a more comprehensive evaluation of the professionals by simultaneously considering the multiple aspects and characteristics. Multimodal evaluations are also more robust to noise or problems in one of the sensors. Furthermore, transfer learning can greatly reduce the data necessary to develop these models and improve their generalization to other contexts. Finally, the interpretability of the AI models is key to providing helpful feedback, with recommendations and suggestions. Consequently, we encourage future researchers to test these techniques in their proposals.

Our major concern is the lack of maturity of the proposals. Most of them are not fully integrated, not tested with end-users, and do not consider ethical and privacy aspects. Typically, the authors' efforts conclude with the development of AI models without advancing to the application of their proposals in real-world settings. In addition, there is no improvement in the maturity of the architectures employed over the publication years. For most authors, no new publications have been found regarding integrating their proposals into final platforms [14, 32, 36, 77]. Consequently, these tools are not readily available to genuinely enhance the training and assessment of professionals. Therefore, the first priority in this field should be to transfer existing proposals to real environments to test and validate the different approaches used. This also means that the adoption by users of these proposals is unknown. Some professionals may not like these platforms, or some changes may be necessary for certain professionals with less technology experience [31]. Other authors may explore these challenges depending on the type of professional and their ethics and privacy concerns. For instance, teachers and military personnel may have different adoption barriers. In addition, it is necessary to evaluate the effectiveness of using these platforms in the long term [92, 94]. Table 4 contains 12 examples of validated platforms for real-world applications. Therefore, we encourage the rest of the researchers to go a step further and transfer their proposals to real-world applications.

In a nutshell, our main recommendation to researchers in this field is to integrate their proposals into platforms that can really support professional training. In addition, it is crucial to validate the different proposals and approaches for each professional and skill. Improving the maturity of the proposals is key since this research field is useless without real-world benefits. Other lower-priority suggestions for advancing the field are exploring other AI algorithms and data fusion techniques. Specifically, we recommend using methods to reduce the data collection and improve the results' expandability. Finally, virtual reality has a huge potential to create more immersive simulations, obtaining more realistic training and evaluations.

6.3 Limitations

This literature review is mainly limited by the paper selection. First of all, we have only analyzed papers that incorporate AI. However, some professionals could be evaluated without the application of AI. Cyber Ranges are platforms designed to train cyber security professionals, but they do not have to include AI [6, 81]. Besides, **Game-based Assessment (GBA)** is another approach to train or assess skills that could be relevant for professionals [41]. On the other hand, we required that the papers include in their title at least one of the terms: skill, competence, capability, capacity, or ability. Nevertheless, some proposals in the field of affective computing could be adapted to train or evaluate some skills, for example, stress management from the emotion recognition field [7].

Another limitation is the academic orientation of the literature review. This academic orientation might not fully cover commercial platforms designed for companies to train and evaluate their workers [67]. Additionally, there is a possibility that certain platforms, especially those developed for military training, are not publicly disclosed. This lack of public availability is often intentional to prevent imitation and to conceal potential vulnerabilities from adversaries. Despite their potential relevance, these platforms remain outside the scope of our review because they are not publicly available [2].

7 Conclusions

This literature review analyzed 83 papers on using AI to support the training and assessment of professionals, following the PRISMA methodology. This systematic review identified a significant interest in specific professionals and skills, particularly healthcare professionals. In addition, most proposals were tailored to evaluate specific skills of some professionals, and only a few studies addressed generic skills applicable to different professions, such as communication and leadership. Regarding technologies utilized, this review literature revealed a scarcity of data fusion and advanced AI technologies. Machine learning, particularly supervised learning, was employed in most papers. Transfer learning and auxiliary supervision could mitigate the extensive data requirements of supervised models. Besides, XAI was also neglected, even though it could significantly enhance the feedback and suggestions provided to the learners. Computer vision was widely used, correlating with the high utilization of cameras in these studies. The preference for cameras is due to they are non-invasive and affordable. However, this leads to a dependence on feature extraction through computer vision techniques. Additionally, the literature review highlighted the potential of virtual reality to create more realistic and immersive simulations. Therefore, authors overlooked more complex techniques that could improve the accuracy and robustness of their proposals.

The literature review also revealed a concerning trend regarding the immaturity of the proposed platforms. Most proposals lack comprehensive architecture, ethical and privacy considerations, and validations with end-users. This worrying finding implies a considerable gap between research and real-world applications. Most authors concentrate on researching the use of AI in this field but do not finally materialize their proposals into platforms that can be utilized. Therefore, most of the platforms do not support the training of professionals. We consider this gap the main

challenge to be faced by future researchers, as it limits the application of the proposed tools in real environments. Addressing this gap is crucial for improving the education of professionals through the feedback and evaluations in hands-on exercises that many professionals demand.

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